Training, test, validation

Split data randomly into training set, test set and validation set.

Training set: use to train the model

validation set: hyperparameter tuning by evaluating performance of the model using different hyperparameter values (model selection)

Test set evaluate the performance of final model (after hyperparameter tuning)

Performance measures of linear regression

Mean absolute error (MAE):

$$MAE = \frac{1}{m} \sum_{i=1}^{m} |y_i - \hat{y}_i|$$

MRSS / Mean squared error

$$MSE = \frac{1}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

Root mean squared error

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2}$$

Grid search of hyperparameter tuning:

1. List of hyperparameter

$$\alpha = \{.01, .001, .1\}$$
$$\rho = \{.001, .0001, .1\}$$

- Create grid of possible values of all hyperparameter

It means all combinations of these values will be checked.

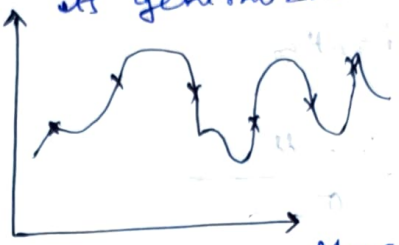
2. Train model using training set.
3. validate it using validation set for each combination.
4. Choose the best hyperparameter combination.
5. Retrain the model with best set.
6. Test with test set.

Overfitting

Model performs well using training samples but does not perform well on the test samples.
→ try to fit with every training samples / hamper its generalization capacity.



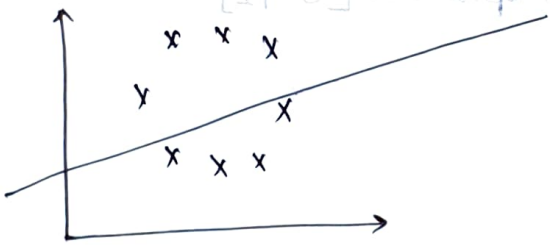
original



Overfitting pass through each points.

Underfitting

→ Model does not perform well on training as well as test sample.
→ Too simple model with complex non-linear data



→ not enough training.

Feature scaling

Different feature scale hamper the learning speed of gradient descent.

Feature 1 → range → 1 to 5

Feature 2 → range → 10,000 to 1,00,000

applied feature column-wise

Min-max scaling

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Standardization

$$x' = \frac{x - \mu}{\sigma}$$

Absolute Max

$$x' = \frac{x}{\max(|x|)}$$

1. Design a linear regression model using batch mode approach by considering following dataset and use this model to predict the output for input pattern [3 12]

X_1	X_2	y
1	2	6
2	10	24

- learning rate = 0.1
- number of training step = 2
- All parameters are initialized as 1

(discussed in class)